DATA SCIENCE INTERNSHIP - WEEK 1

Task 2: Sales Data Analysis

Business Performance, Sales Trends and Customer Insights
Organization: WeIntern Pvt Ltd
Tool: Python / Pandas / Jupyter Notebook

1. Objective

Task 2 analyzes the Sample Superstore sales dataset to evaluate business performance, identify sales and customer patterns, and generate evidence-based insights. The analysis emphasizes accurate KPI definitions, unique-order logic, trend interpretation, category and product performance, regional differences, and customer purchasing behavior.

2. Dataset Overview and Analytical Preparation

Item	Result
Source dataset	Sample - Superstore
Order-line records	9,994
Unique orders	5,009
Unique customers	793
Unique products	1,850
Fully duplicated rows	0
Invalid order dates	0
Invalid ship dates	0

The dataset is stored at order-line level, so a single Order ID may appear on multiple rows. Repeated Order IDs are valid line items rather than automatic duplicates. Total Orders therefore uses distinct Order IDs. The uploaded CSV contains both hyphen- and slash-separated U.S. date strings; delimiters were normalized before parsing as month/day/year. Column names were standardized to lowercase snake_case and year/month fields were derived for trend analysis.

3. KPI Summary

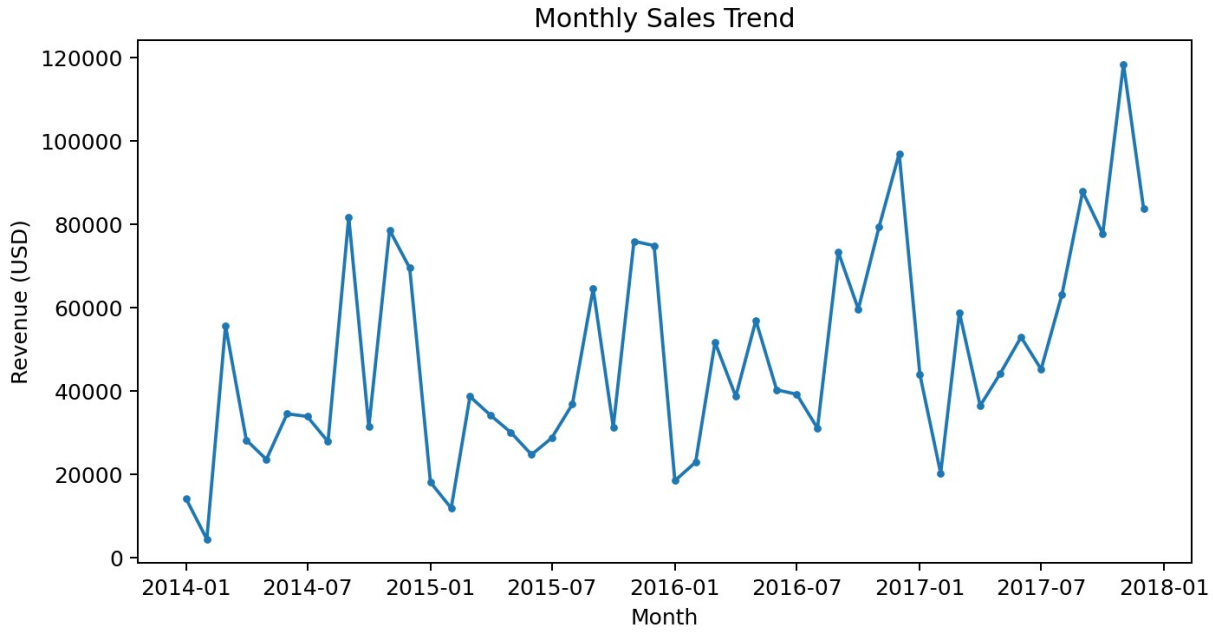
KPI	Definition	Result
Total Revenue	Sum of Sales	$2,297,200.86
Total Orders	Distinct Order ID	5,009
Average Order Value	Revenue / distinct orders	$458.61
Total Units Sold	Sum of Quantity	37,873
Top-Selling Category	Highest category revenue	Technology
Top Revenue-Generating Product	Highest product revenue	Canon imageCLASS 2200 Advanced Copier

Technology generated $836,154.03 (36.40% of revenue). Canon imageCLASS 2200 Advanced Copier generated $61,599.82.

4. Sales Trend Analysis

Year	Revenue	Orders	Units	YoY Revenue Growth
2014	$484,247.50	969	7,581	-
2015	$470,532.51	1,038	7,979	-2.83%
2016	$609,205.60	1,315	9,837	29.47%
2017	$733,215.26	1,687	12,476	20.36%

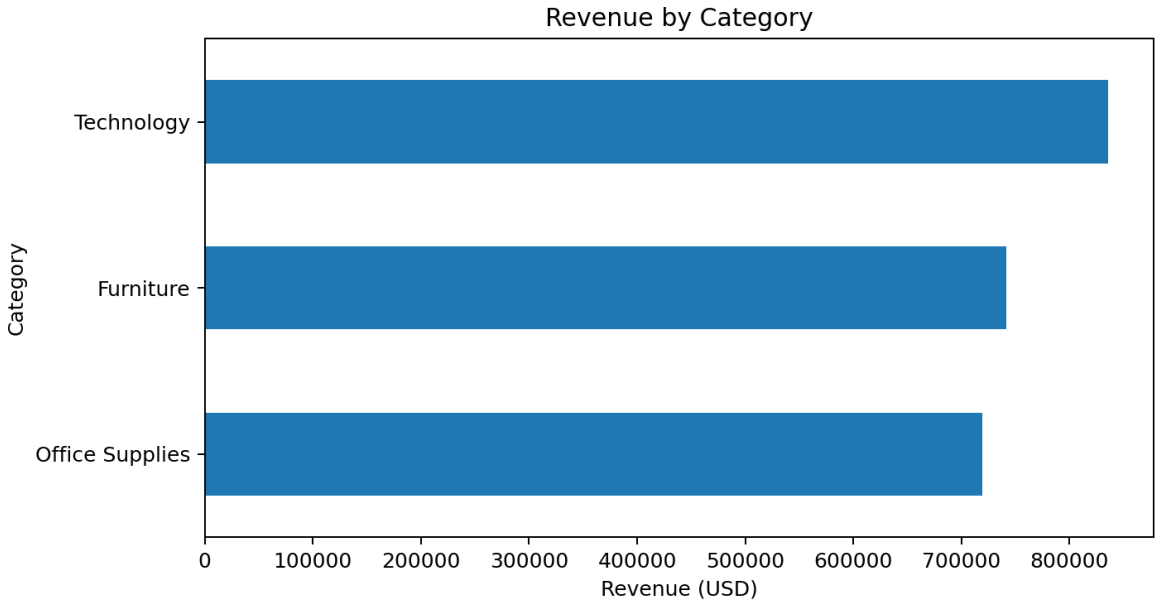
Revenue declined by 2.83% in 2015, then increased by 29.47% in 2016 and 20.36% in 2017. Orders and units also increased substantially by the end of the period. The dataset supports the observed trend but does not establish the external causes of those changes.



Highest month: 2017-11 with $118,447.82, 261 orders and 1,840 units. Lowest month: 2014-02 with $4,519.89, 28 orders and 159 units.

5. Category and Product Performance

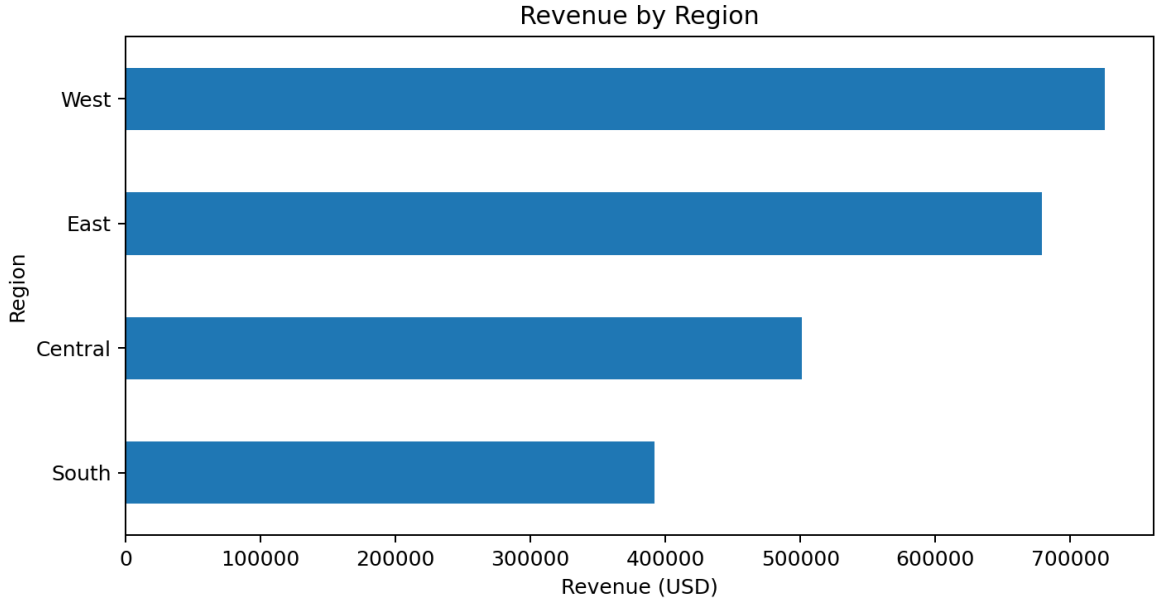
Category	Revenue	Revenue Share
Technology	$836,154.03	36.40%
Furniture	$741,999.80	32.30%
Office Supplies	$719,047.03	31.30%



The highest-revenue product was Canon imageCLASS 2200 Advanced Copier at $61,599.82 from 20 units. The highest-volume product was Staples at 215 units. Revenue leadership and volume leadership therefore differ.

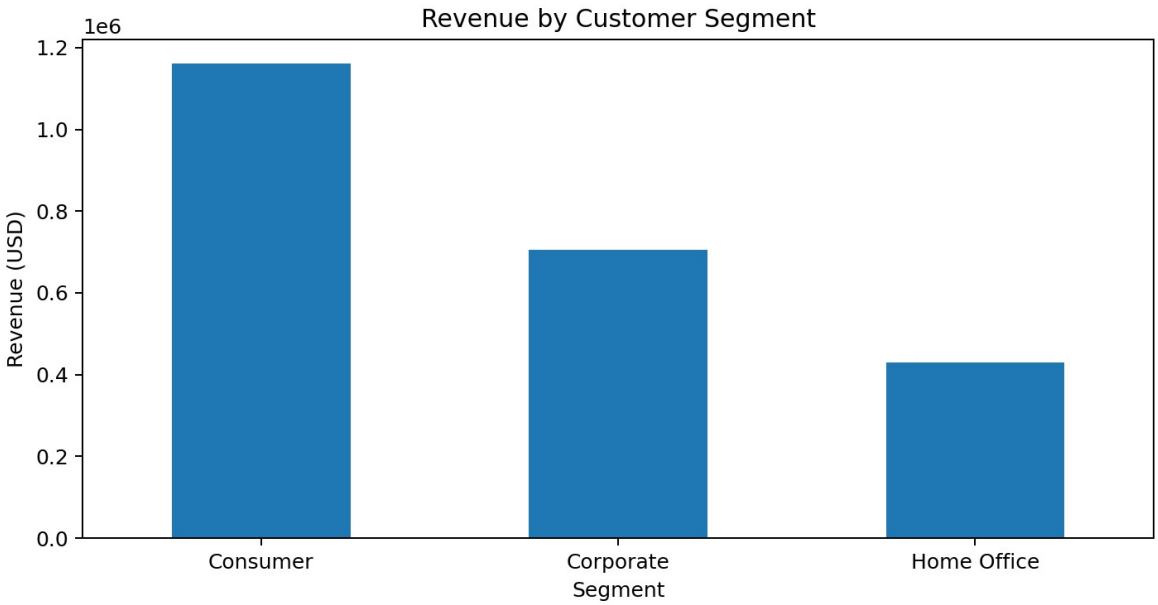
6. Regional and Segment Analysis

Region	Revenue	Revenue Share	Orders
West	$725,457.82	31.58%	1,611
East	$678,781.24	29.55%	1,401
Central	$501,239.89	21.82%	1,175
South	$391,721.91	17.05%	822



Segment	Revenue	Revenue Share	Orders	Customers
Consumer	$1,161,401.34	50.56%	2,586	409

Corporate	$706,146.37	30.74%	1,514	236
Home Office	$429,653.15	18.70%	909	148



7. Customer Insights

Customer Measure	Result
Total customers	793
Repeat customers	781
One-time customers	12
Repeat customer rate	98.49%
Repeat customer revenue share	99.78%
Highest-revenue customer	Sean Miller - $25,043.05 (5 orders)
Most-frequent customer	Emily Phan - 17 orders ($5,478.06)

Repeat customers account for nearly all observed revenue in this dataset. This is a descriptive pattern, not evidence that repeat purchasing caused higher revenue. Customer value also differs from purchase frequency: the highest-revenue customer is not the customer with the most orders.

8. Required Written Observations

1. The dataset contains 5,009 unique orders generating $2,297,200.86 from 37,873 units, with an average order value of $458.61.

2. Revenue declined 2.83% in 2015, then increased 29.47% in 2016 and 20.36% in 2017, alongside higher order counts and units sold.

3. November 2017 was the highest-revenue month at $118,447.82 from 261 orders and 1,840 units.

4. Technology led category revenue at $836,154.03, or 36.40% of total revenue.

5. Canon imageCLASS 2200 Advanced Copier led product revenue at $61,599.82, while Staples led unit volume at 215 units.

6. The West led regional revenue at $725,457.82, or 31.58% of total revenue.

7. 781 of 793 customers (98.49%) were repeat customers and generated 99.78% of observed revenue.

8. Sean Miller generated the most customer revenue ($25,043.05), while Emily Phan placed the most orders (17).

9. Recommendations

- Track revenue, order count, units, and average order value together so growth is not interpreted from a single metric.
- Evaluate high-revenue products and high-volume products separately because they represent different commercial patterns.
- Use regional and segment contribution as inputs for deeper profitability and resource-allocation analysis rather than relying on revenue alone.
- Retain repeat-customer and customer-value measures in future reporting to distinguish frequency from monetary contribution.

10. Limitations and Assumptions

- The source data is at order-line level; distinct Order ID is used for order-based metrics.
- Revenue is calculated from the provided Sales field.
- Repeat customers are defined as customers with more than one distinct order in the supplied dataset.
- The analysis describes associations and historical patterns; it does not establish causal explanations.
- Monthly and yearly differences may reflect factors that are not represented in the available variables.
- Findings apply to the time period and customers represented in the supplied Sample Superstore dataset.

11. Deliverables Coverage

Task 2 Expected Deliverable	Where Covered
KPI summary table	Section 3
5-8 written observations	Section 8 - eight evidence-based observations
At least one trend-based interpretation	Section 4
At least one customer insight section	Section 7
Clean analytical notebook/script	Task2.ipynb (repository deliverable)
Clear grouping logic / KPI definitions	Sections 2-3
Observations separated from recommendations	Sections 8 and 9

This report accompanies the analytical notebook. The notebook remains the executable source of the calculations, while this report presents the required results and interpretations in a review-ready format.

12. Conclusion

Task 2 demonstrates a complete sales-analysis workflow built on distinct-order logic and evidence-based interpretation. The analysis establishes the core sales KPIs, identifies time, category, product, regional and segment patterns, and provides a dedicated customer analysis without attributing unsupported causes to the observed results.